Rank-27 Compact Definition Core and Vivify Multiset Closure

Tactix engineering handoff

24 August 2026

Executive outcome

The latest cumulative Tactix pin keeps the accepted Rank-27 search trajectory and cuts checked wall time from 579.772 to 504.297 seconds. BVE falls from 174.204 to 94.389 seconds. The repaired 1.633-GB DRAT certificate verifies and is deleted, but the result remains 1.681 times the 300-second objective and 1.76 times the 286.30-second current-Kissat diagnostic control.

The improvement comes from general representation fidelity to Kitten. BVE's definition variables are already dense in first-import order, so the shared subsolver no longer rebuilds an identical outer map. It stores only the watched occurrence structure that Kitten's model-flip path reads. Finally, second-core membership uses persistent scratch and retained original clauses compact in place, matching `kitten_shrink_to_clausal_core` without changing Tactix's proof-visible clause header. No benchmark identity, rank, hash, deadline, per-instance arm, or tuned constant enters the solver.

The apparent next semantic lead in conflict-9,449 Vivify is now closed as a false positive. Candidate order, decisions, and normalized logical ticks are exact over 4,915 paired candidates. Full binary-multiset snapshots show that both solvers derive the same 19 binaries. Tactix's scalar count ends one lower only because it enters with two copies of one pair and normalizes to one, whereas Kissat enters and exits with one. No Vivify repair is justified.

Frozen authorities and checked Rank-27 witness

The immutable input is `benchmarks/cache/tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. No frozen scoring reference was rerun. The current-Kissat 286.30-second value comes from a separately pinned diagnostic build at the latest common semantic state.

Checked campaign `benchmarks/runs/20260824-0306-tierd-rank27-bve-definition-compact-kitten-witness/` returns UNSAT in 503.787 internal and 504.297 wall seconds with peak solver RSS 654,816 KiB. It preserves exactly 3,747,797 conflicts, 177,941,301 decisions, 61,793,863 BVE resolutions, 1,223,465 definition checks, 2,505,837,432 definition ticks, and 352,186 eliminated variables. Its 1,633,208,191-byte proof has SHA-256 `7a9c2c04c8dc143ae498eb972ec403566cb9287d29adf8559b77a2aeb5cf1885`; `drat-trim` verifies it in 106.339 seconds. The proof, checker products, and decompressed input are deleted, leaving zero certificate residue.

Compact-core representation repair

Three nested copies were removed without changing logical ownership. `reset_compact_kitten` consumes BVE's existing dense first-occurrence image directly. Fixed-mode-only all-clause occurrence lists are absent in the Kitten path because upstream Kitten's flip predicate scans watched clauses.

The second-core pass retains a parallel membership byte—Tactix's two-word proof header has no spare flag—and compacts original clauses in the existing arena instead of allocating a second arena.

The complete mp1 observer remains byte-exact over all 16,395 paired definition records, including candidate and clause import order, variable/side counts, both core size/split/hash values, cumulative ticks, allowance, generator, and terminal outcome. Thus these changes are engine representation, not a new BVE proof rule or a policy shortcut.

The exact-source B28 plus fixed mechanism-relevant C5 guard is `benchmarks/runs/20260824-0319-BC-bve-definition-inplace-core-guard/`. All 66 candidate/comparator tasks validate: each arm has 17 SAT models and 16 UNSAT proofs, with no invalid, missing, regression, cleanup, or residue entry. Candidate PAR-2 is 1,640.738 seconds versus 2,223.039 seconds shipped, a 0.738061 ratio. The summary SHA-256 is `747bddf6bd507dd76aa1e2d95cb5e161a146b090ad1ea2d9ec6cf3c7b6cccc22`.

Why the Vivify count was not a pruning loss

Macro and rule-boundary traces agree through the second conflict-9,449 Vivify entry. Tactix exits with 753,322 binary copies and Kissat with 753,323, while both have 574,136 irredundant and 1,902 redundant large clauses. A scalar reading therefore suggested that Tactix learned one fewer binary.

Campaign `benchmarks/runs/20260824-0323-tierd-rank27-current-vivify9449-binary-sets/` enumerates sorted permanent-external-literal multisets at every Vivify boundary. During the second invocation each solver adds the exact same 19 copies. Kissat deletes 11 and Tactix deletes 12. The sole extra Tactix deletion is pair $(-354369, -354368)$: Tactix has two copies at entry and one at exit; Kissat has one at both boundaries. The two terminal multisets retain a separate, inherited equivalent-representative difference, but the Vivify delta contains no lost consequence. This demonstrates why worse-first divergence analysis must compare exact deltas, not only populations.

The diagnostic snapshot is disabled by default and exists only in event-trace builds. It retains duplicate copies, sorts by permanent external literals, and changes neither proof rules nor scheduling. Its Tactix path has a unit test; the Kissat overlay has a static construction test. Both diagnostic jobs complete in under 47 seconds and leave no staged input or certificate.

Remaining worse-side throughput owner

The 300-second objective is still open. Current Tactix and current Kissat reach the same terminal Rank-27 search trajectory, yet Tactix needs 503.787 seconds internally versus 286.30 seconds. Elimination alone is 94.389 versus 16.32 seconds. This is the preferred divergence class: the same useful work is executed much more slowly.

The complete terminal-BVE observers give a sharper next lead. In that phase, Kissat has 292,771 definition attempts and 18,112,153 terminal embedded ticks; Tactix has 369,081 attempts and 280,614,670 ticks. Tactix has 32,932 attempts whose imported variable image exceeds 1,024 variables, accounting for 249,343,427 ticks; Kissat has none in that band. The first Tactix attempt above 256 variables is external candidate 396947: four long occurrence clauses span 666 variables and cost 674 embedded ticks, while current Kissat declines the same candidate without a definition attempt. Because the formula states have already diverged, this is a localization lead, not permission to patch BVE admission. The next trace must identify the upstream clause/representation transaction that leaves Tactix with these long pivot clauses.

Correctness and operational state

The cumulative Release suite is 73/74: every executable, proof, configuration, and hygiene test passes; only the known documentation-size policy fails because the living progress and implementation notes exceed their advisory limits. Repository authority forbids silent compaction, so no history was deleted. Focused event-trace tests pass, and `git diff -check` is clean.

The exact-source native-LRAT fuzz campaign `benchmarks/runs/20260824-0318-bve-definition-inplace-core-fuzz/` passes 10,000 cases with zero failures: 5,188 SAT plus 3,576 UNSAT terminating checked verdicts, 1,236 explicit five-second non-terminations, and 1,459,984 stress-target applications. Twenty-five bounded timeout CNFs are retained as diagnostic inputs, while no proof or checker artifact remains. No unit or fuzz soundness/crash trigger required the authorized Claude repair path.

The prior full Tier-D comparison remains independent and continues without blocking Rank-27 work. After the fuzz gate, the current Release binary is frozen as an independent read-only campaign copy for a new SC24–26 union run under 1,800 seconds, 16 GiB, two CPUs per task, and array throttle 85. Its runner validates every SAT model and UNSAT certificate, deletes all certificate artifacts, and aggregates asynchronously; Rank-27 work continues rather than waiting for that array.

Clean handoff boundary

The current interactive allocation expires roughly four hours after this checkpoint. A safe handoff is: finish fuzz; freeze and submit the current Tier-D campaign; run the checked current Rank-27 witness; record whether in-place core compaction reduces the 249-million-tick large-image tail; and only then open the selected long-clause lineage trace. Commit and push the source, tests, documentation, and this PDF with a clean synchronized worktree. Do not reopen Vivify from its scalar binary count and do not tune a candidate admission threshold to external 396947.